Pregunta 1

a) $I = \displaystyle\int \frac{(x-3)dx}{(2x^2-10x+13)\sqrt{10x^2-42x+45}}$

Realizamos la sustitución de la inversa para eliminar el término lineal del numerador:

$$x-3=\frac{1}{t} \implies x=\frac{1}{t}+3 \implies dx=-\frac{1}{t^2}dt$$

Transformamos los componentes del denominador en términos de t:

$$2x^2-10x+13=2\left(\frac{1}{t}+3\right)^2-10\left(\frac{1}{t}+3\right)+13=\frac{(t+1)^2+1}{t^2}$$
$$10x^2-42x+45=10\left(\frac{1}{t}+3\right)^2-42\left(\frac{1}{t}+3\right)+45=\frac{9(t+1)^2+1}{t^2}$$

Reemplazamos estos resultados en la integral original, donde las potencias de t se cancelan perfectamente:

$$I=\int \frac{\left(\frac{1}{t}\right)\left(-\frac{1}{t^2}\right)dt}{\left(\frac{(t+1)^2+1}{t^2}\right)\frac{\sqrt{9(t+1)^2+1}}{t}}=\int \frac{-dt}{\left[(t+1)^2+1\right]\sqrt{9(t+1)^2+1}}$$

Hacemos una sustitución de traslación: $u=t+1 \implies du=dt$:

$$I=\int \frac{-du}{(u^2+1)\sqrt{9u^2+1}}$$

Aplicamos la sustitución trigonométrica $3u=\tan(\theta) \implies du=\frac{1}{3}\sec^2(\theta)d\theta$:

$$I=\int \frac{-\frac{1}{3}\sec^2(\theta)d\theta}{\left(\frac{\tan^2(\theta)}{9}+1\right)\sec(\theta)}=\int \frac{-3\cos(\theta)d\theta}{9-8\text{sen}^2(\theta)}$$

Utilizamos la sustitución inmediata $w=\text{sen}(\theta) \implies dw=\cos(\theta)d\theta$:

$$I=\int \frac{-3dw}{9-8w^2}=\frac{3}{8}\int \frac{dw}{w^2-\frac{9}{8}}$$

Integrando mediante fracciones parciales estándar:

$$I=\frac{3}{8}\cdot\frac{2\sqrt{2}}{6}\ln\left|\frac{w-\frac{3}{2\sqrt{2}}}{w+\frac{3}{2\sqrt{2}}}\right|+C$$

Regresando de forma sucesiva a la variable original x:

$$\boxed{I=\frac{\sqrt{2}}{8}\ln\left|\frac{2\sqrt{2}w-3}{2\sqrt{2}w+3}\right|+C}$$

Donde:

$$\boxed{w=\frac{3(x-2)}{\sqrt{10x^2-42x+45}}}$$

b) $I = \displaystyle\int \frac{(\operatorname{sen}(x) + x\cos(x))dx}{(1 - x\operatorname{sen}(x))\sqrt{-1 + x^2 - x^2\cos^2(x)}}$

Identidad en el radicando:

$$-1 + x^2(1 - \cos^2(x)) = -1 + x^2\operatorname{sen}^2(x) = (x\operatorname{sen}(x))^2 - 1$$

$$I = \int \frac{(\operatorname{sen}(x) + x\cos(x))dx}{(1 - x\operatorname{sen}(x))\sqrt{(x\operatorname{sen}(x))^2 - 1}}$$

Sustitución $u = x\operatorname{sen}(x) \implies du = (\operatorname{sen}(x) + x\cos(x))dx$:

$$I = \int \frac{du}{(1-u)\sqrt{u^2-1}} = \int \frac{-du}{(u-1)\sqrt{u^2-1}}$$

Sustitución $u - 1 = \frac{1}{t} \implies u = \frac{1}{t} + 1 \implies du = -\frac{1}{t^2}dt$:

$$\sqrt{u^2-1} = \sqrt{\left(\frac{1}{t}+1\right)^2 - 1} = \sqrt{\frac{1}{t^2} + \frac{2}{t}} = \frac{\sqrt{1+2t}}{t}$$

$$I = \int \frac{-\left(-\frac{1}{t^2}\right)dt}{\left(\frac{1}{t}\right)\left(\frac{\sqrt{1+2t}}{t}\right)} = \int \frac{\frac{1}{t^2}dt}{\frac{\sqrt{1+2t}}{t^2}}$$

$$I = \int (1+2t)^{-1/2}dt = \frac{(1+2t)^{1/2}}{2(1/2)} + C = \sqrt{1+2t} + C$$

Restitución de variables:

$$t = \frac{1}{u-1} = \frac{1}{x\operatorname{sen}(x) - 1} \implies 1 + 2t = 1 + \frac{2}{x\operatorname{sen}(x) - 1} = \frac{x\operatorname{sen}(x) + 1}{x\operatorname{sen}(x) - 1}$$

$$\boxed{I = \sqrt{\frac{x\operatorname{sen}(x) + 1}{x\operatorname{sen}(x) - 1}} + C}$$

c) $I = \displaystyle\int \frac{\cosh(x)dx}{(1+\operatorname{senh}(x))^5(2+\operatorname{senh}(x))^6}$

Sustitución $u = \operatorname{senh}(x) \implies du = \cosh(x)dx$:

$$I = \int \frac{du}{(u+1)^5(u+2)^6}$$

Sustitución homográfica:

$$\begin{aligned} t = \frac{u+1}{u+2} &\implies t(u+2) = u+1 \implies ut + 2t = u+1 \\ &\implies u(t-1) = 1-2t \implies u = \frac{1-2t}{t-1} = \frac{2t-1}{1-t} \end{aligned}$$

Diferencial:

$$du = \frac{2(1-t)-(2t-1)(-1)}{(1-t)^2}dt = \frac{2-2t+2t-1}{(1-t)^2}dt = \frac{1}{(1-t)^2}dt$$

Transformación de los factores:

$$\begin{aligned} u+1 &= \frac{2t-1}{1-t} + 1 = \frac{2t-1+1-t}{1-t} = \frac{t}{1-t} \\ u+2 &= \frac{2t-1}{1-t} + 2 = \frac{2t-1+2-2t}{1-t} = \frac{1}{1-t} \end{aligned}$$

Sustitución en la integral:

$$\begin{aligned} I &= \int \frac{\frac{1}{(1-t)^2}dt}{\left(\frac{t}{1-t}\right)^5\left(\frac{1}{1-t}\right)^6} = \int \frac{\frac{1}{(1-t)^2}dt}{\frac{t^5}{(1-t)^{11}}} \\ I &= \int \frac{(1-t)^{11}}{t^5(1-t)^2}dt = \int \frac{(1-t)^9}{t^5}dt \end{aligned}$$

Binomio de Newton:

$$\begin{aligned} (1-t)^9 &= \sum_{k=0}^{9} \binom{9}{k}(-t)^k \\ &= 1 - 9t + 36t^2 - 84t^3 + 126t^4 - 126t^5 + 84t^6 - 36t^7 + 9t^8 - t^9 \end{aligned}$$

Integración término a término:

$$I = \int \left(t^{-5} - 9t^{-4} + 36t^{-3} - 84t^{-2} + 126t^{-1} - 126 + 84t - 36t^2 + 9t^3 - t^4\right) dt$$

$$I = \frac{t^{-4}}{-4} - 9\frac{t^{-3}}{-3} + 36\frac{t^{-2}}{-2} - 84\frac{t^{-1}}{-1} + 126\ln|t| - 126t + 84\frac{t^2}{2} - 36\frac{t^3}{3} + 9\frac{t^4}{4} - \frac{t^5}{5} + C$$

$$\boxed{I = -\frac{1}{4t^4} + \frac{3}{t^3} - \frac{18}{t^2} + \frac{84}{t} + 126\ln|t| - 126t + 42t^2 - 12t^3 + \frac{9t^4}{4} - \frac{t^5}{5} + C}$$

Donde:

$$\boxed{t = \frac{\operatorname{senh}(x)+1}{\operatorname{senh}(x)+2}}$$

d) $I = \displaystyle\int \frac{2dx}{\operatorname{sen}(2x)\sqrt[3]{1+\sqrt[4]{\tan^3(x)}}}$

Sustitución $y = \tan(x) \implies \operatorname{sen}(2x) = \frac{2y}{1+y^2}, \quad dx = \frac{dy}{1+y^2}$:

$$I = \int \frac{2\left(\frac{dy}{1+y^2}\right)}{\left(\frac{2y}{1+y^2}\right)\sqrt[3]{1+y^{3/4}}} = \int \frac{dy}{y\sqrt[3]{1+y^{3/4}}}$$

Sustitución para eliminar el radical:

$$z^3 = 1 + y^{3/4} \implies y^{3/4} = z^3 - 1 \implies y = (z^3-1)^{4/3}$$

$$dy = \frac{4}{3}(z^3-1)^{1/3} \cdot 3z^2 dz = 4z^2(z^3-1)^{1/3}dz$$

$$I = \int \frac{4z^2(z^3-1)^{1/3}dz}{(z^3-1)^{4/3}\cdot z} = \int \frac{4z^2}{z(z^3-1)}dz = \int \frac{4z}{z^3-1}dz$$

$$I = \int \frac{4z}{(z-1)(z^2+z+1)}dz$$

Fracciones parciales:

$$\frac{4z}{(z-1)(z^2+z+1)} = \frac{A}{z-1} + \frac{Bz+C}{z^2+z+1}$$

$$4z = A(z^2+z+1) + (Bz+C)(z-1)$$

$$Si\ z = 1 \implies 4 = A(1+1+1) \implies \boxed{A = 4/3}$$

$$Si\ z = 0 \implies 0 = A(1) + C(-1) \implies \boxed{C = 4/3}$$

$$Coeficiente\ z^2 \implies 0 = A + B \implies \boxed{B = -4/3}$$

Resolviendo la integral descompuesta:

$$I = \int \frac{4/3}{z-1}dz + \int \frac{-\frac{4}{3}z + \frac{4}{3}}{z^2+z+1}dz$$

$$I = \frac{4}{3}\ln|z-1| - \frac{4}{3}\int \frac{\frac{1}{2}(2z+1) - \frac{3}{2}}{z^2+z+1}dz$$

$$I = \frac{4}{3}\ln|z-1| - \frac{4}{3}\left[\frac{1}{2}\int \frac{2z+1}{z^2+z+1}dz - \frac{3}{2}\int \frac{dz}{\left(z+\frac{1}{2}\right)^2 + \frac{3}{4}}\right]$$

$$I = \frac{4}{3}\ln|z-1| - \frac{2}{3}\ln(z^2+z+1) + 2\int \frac{dz}{\left(z+\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$I = \frac{4}{3}\ln|z-1| - \frac{2}{3}\ln(z^2+z+1) + 2\left(\frac{1}{\sqrt{3}/2}\right)\arctan\left(\frac{z+1/2}{\sqrt{3}/2}\right)$$

$$\boxed{I = \frac{4}{3}\ln|z-1| - \frac{2}{3}\ln(z^2+z+1) + \frac{4}{\sqrt{3}}\arctan\left(\frac{2z+1}{\sqrt{3}}\right) + C}$$

Donde:

$$\boxed{z = \sqrt[3]{1+\tan^{3/4}(x)}}$$

Pregunta 2

Demostrar que $\displaystyle\int \csc^n(x)dx = \frac{-1}{n-1}\cot(x)\csc^{n-2}(x) + \frac{n-2}{n-1}\int \csc^{n-2}(x)dx$
Separamos el integrando:

$$I_n = \int \csc^{n-2}(x)\csc^2(x)dx$$

Integración por partes:

$$\begin{aligned} u = \csc^{n-2}(x) &\implies du = -(n-2)\csc^{n-2}(x)\cot(x)dx \\ dv = \csc^2(x)dx &\implies v = -\cot(x) \end{aligned}$$

Aplicando $\int udv = uv - \int vdu$:

$$I_n = -\cot(x)\csc^{n-2}(x) - (n-2)\int \csc^{n-2}(x)\cot^2(x)dx$$

Identidad pitagórica $\cot^2(x) = \csc^2(x) - 1$:

$$I_n = -\cot(x)\csc^{n-2}(x) - (n-2)I_n + (n-2)I_{n-2}$$

Despejando I_n:

$$\boxed{I_n = \frac{-1}{n-1}\cot(x)\csc^{n-2}(x) + \frac{n-2}{n-1}I_{n-2}} \quad \text{(L.Q.Q.D)}$$

Cálculo de $\int \csc^5(x)dx$:
Para $n = 5$, $n = 3$ y $n = 1$:

$$I_5 = -\frac{1}{4}\cot(x)\csc^3(x) + \frac{3}{4}I_3$$

$$I_3 = -\frac{1}{2}\cot(x)\csc(x) + \frac{1}{2}I_1$$

$$I_1 = \int \csc(x)dx = \ln|\csc(x) - \cot(x)|$$

Sustituyendo I_1 en I_3, y luego I_3 en I_5:

$$I_5 = -\frac{1}{4}\cot(x)\csc^3(x) + \frac{3}{4}\left[-\frac{1}{2}\cot(x)\csc(x) + \frac{1}{2}\ln|\csc(x) - \cot(x)|\right]$$

$$\boxed{I_5 = -\frac{1}{4}\cot(x)\csc^3(x) - \frac{3}{8}\cot(x)\csc(x) + \frac{3}{8}\ln|\csc(x) - \cot(x)| + C}$$

Pregunta 3

Exprese como una integral definida $\displaystyle\sum_{i=1}^{n} \frac{n^2+8ni+16i^2}{n^3+6n^2i+24ni^2+32i^3}$

Dividimos numerador y denominador entre la máxima potencia del parámetro n^3:

$$S_n = \sum_{i=1}^{n} \frac{\frac{n^2}{n^3}+\frac{8ni}{n^3}+\frac{16i^2}{n^3}}{\frac{n^3}{n^3}+\frac{6n^2i}{n^3}+\frac{24ni^2}{n^3}+\frac{32i^3}{n^3}}$$
$$S_n = \sum_{i=1}^{n} \frac{\frac{1}{n}+8\left(\frac{1}{n}\right)\left(\frac{i}{n}\right)+16\left(\frac{1}{n}\right)\left(\frac{i}{n}\right)^2}{1+6\left(\frac{i}{n}\right)+24\left(\frac{i}{n}\right)^2+32\left(\frac{i}{n}\right)^3}$$

Factorizamos el diferencial común $\frac{1}{n}$:

$$S_n = \sum_{i=1}^{n} \frac{1}{n}\left[\frac{1+8\left(\frac{i}{n}\right)+16\left(\frac{i}{n}\right)^2}{1+6\left(\frac{i}{n}\right)+24\left(\frac{i}{n}\right)^2+32\left(\frac{i}{n}\right)^3}\right]$$

Identificamos los elementos de la Suma de Riemann con $\Delta x = \frac{1}{n}$ y $x_i = \frac{i}{n}$ acotados en $[0,1]$:

$$\begin{aligned}\lim_{n\to\infty} S_n &= \int_0^1 \frac{1+8x+16x^2}{1+6x+24x^2+32x^3}dx \\ &= \int_0^1 \frac{(1+4x)^2}{1+6x+24x^2+32x^3}dx\end{aligned}$$

$$\boxed{\int_0^1 \frac{1+8x+16x^2}{1+6x+24x^2+32x^3}dx}$$

Pregunta 4

Mediante sumas superiores e inferiores, halle el valor aproximado de $\int_{-3}^{2} \frac{x}{1+x^2}dx$, partición $P = \{-3, -2, -1, 0, 1, 2\}$. Dar una cota superior para el error.

Derivada de la función $f(x) = \frac{x}{1+x^2}$:

$$f'(x) = \frac{(1)(1+x^2) - x(2x)}{(1+x^2)^2}$$
$$f'(x) = \frac{1+x^2-2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

Puntos críticos:

$$1 - x^2 = 0 \implies x = \pm 1$$

Evaluación de los puntos en la partición (longitud de subintervalo $\Delta x_i = 1$):

- $f(-3) = \frac{-3}{1+(-3)^2} = -0{,}3$
- $f(-2) = \frac{-2}{1+(-2)^2} = -0{,}4$
- $f(-1) = \frac{-1}{1+(-1)^2} = -0{,}5$
- $f(0) = \frac{0}{1+0^2} = 0$
- $f(1) = \frac{1}{1+1^2} = 0{,}5$
- $f(2) = \frac{2}{1+2^2} = 0{,}4$

Suma Superior e Inferior:

$$U(P,f) = \sum_{i=1}^{5} M_i \Delta x_i = (-0{,}3) + (-0{,}4) + 0 + 0{,}5 + 0{,}5 = 0{,}3$$
$$L(P,f) = \sum_{i=1}^{5} m_i \Delta x_i = (-0{,}4) + (-0{,}5) + (-0{,}5) + 0 + 0{,}4 = -1{,}0$$

Valor Aproximado:

$$\text{Valor Aproximado} = \frac{U(P,f) + L(P,f)}{2} = \frac{0{,}3 + (-1{,}0)}{2} = \frac{-0{,}7}{2} = -0{,}35$$

Cota Superior del Error:

$$\text{Cota de Error} = U(P,f) - L(P,f) = 0{,}3 - (-1{,}0) = 1{,}3$$

$$\boxed{\text{Valor Aproximado} = -0{,}35}$$

$$\boxed{\text{Cota de Error} = 1{,}3}$$